

Shikha Gupta

Senior Software Development Engineer | Distributed Systems | Sustainability Technology | Application Security | Generative AI

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Profile

I am a software engineering leader with more than 14 years of experience building secure, scalable, and intelligent systems across application security, distributed systems, sustainability technology, and generative AI. At Amazon's Worldwide Sustainability (WWS) organization, I lead technical strategy and architecture for mission-critical sustainability platforms that support carbon accounting, environmental data management, and public carbon footprint disclosures. Earlier at Infosys Research Labs, I co-invented five U.S. patents spanning tokenization, data masking, digital rights management, and electronic signature verification. My work is defined by taking ambiguous, technically complex problems and converting them into durable production systems with measurable business impact, strong security posture, and long-term architectural value.

Areas of Expertise

- Architecture and development of distributed systems and microservices at enterprise scale
- Sustainability data platforms, carbon accounting systems, and life cycle assessment infrastructure
- Generative AI, retrieval-augmented generation (RAG), vector search, and scientific data mapping workflows
- Application security, threat modeling, secure SDLC, data privacy, tokenization, and digital rights management
- AWS technologies including Lambda, Fargate, Step Functions, DynamoDB, DocumentDB, S3, CDK, Bedrock, and OpenSearch
- Cross-functional technical leadership, architecture reviews, vision documents, mentoring, and engineering excellence programs

Education

2015	Master of Science, Computer Science (GPA: 4.0/4.0) University of Southern California, Los Angeles, CA Aug 2014 - Dec 2015
2010	Bachelor of Technology, Computer Science Engineering (Aggregate: 83.76%) Guru Gobind Singh Indraprastha University, Delhi, India Aug 2006 - May 2010

Professional Experience

Amazon.com, Inc. - Seattle / Kirkland, WA, USA

Senior Software Development Engineer, Apr 2023 - Present | Software Development Engineer, Feb 2016 - Mar 2023

I lead technical strategy, software architecture, and secure delivery for sustainability technology systems in Amazon's Worldwide Sustainability organization. As a Technical Bar Raiser, I influence cross-organizational design quality, review more than 20 designs and operational plans annually, and work across engineering, science, and product teams. My responsibilities include long-term technical planning, hands-on system design, application security leadership, mentoring, and driving major greenfield systems from concept to production.

- Since 2020, I have led or co-led more than 15 mission-critical sustainability systems developed with 10+ scientists and researchers and used in Amazon's annual carbon footprint disclosures.
- In Seller Partners Pricing and Fees, I led critical pricing platforms operating in a high-scale environment driven by price-change events and workloads around 40,000 transactions per second.

- I serve as an AppSec Guardian / Security Guardian for the organization and founded the AppSec Samurai initiative to streamline reviews and improve secure development practices.
- I mentor engineers internally and externally on system design, technical leadership, and career growth.

University of Southern California - Los Angeles, CA, USA

Course Producer, Jan 2015 - Dec 2015

Created project assignments, graded exams, and held office hours for the master's course CSCI 571: Web Technologies under Professor Marco Papa, helping support the learning experience for graduate students in applied web engineering.

Infosys Ltd. - Research Labs - Hyderabad, India

Technology Analyst, Oct 2010 - Jul 2014

Developed application-security solutions in digital rights management, data masking, tokenization, and electronic signature verification using cryptography, digital signatures, encryption, X.509 certificates, and PKCS standards. My work combined product engineering with applied research and resulted in five granted U.S. patents later recognized for industry relevance and citation impact.

Hewlett-Packard - Gurgaon, India

Technical Intern, Sep 2009 - Nov 2009

Developed a J2ME mobile application using REST APIs and Symbian S60 SDK for HP printers, with a focus on mobile-device interaction for enterprise hardware workflows.

Technical Leadership and Innovation

2020-2021 | Sustainability Fact Store and SCFS Vision - Amazon Worldwide Sustainability

From 2020 to 2021, I co-created Amazon's centralized Sustainability Fact Store and later authored the three-year vision for the Sustainability Core Fact Store (SCFS), the system of record for sustainability facts ranging from emission factors to grid datasets. The platform was designed to ingest and govern heterogeneous scientific datasets with flexible schemas, fine-grained access control, versioning, and usage visibility. By solving fragmentation across teams and data providers, it established a trustworthy and auditable foundation for carbon footprint calculations. Today SCFS hosts more than 200 scientific datasets and serves more than 20 internal organizations. The roadmap I authored aligned the platform with Amazon's global sustainability priorities and introduced long-term investments in discoverability, traceability, governance, and advanced analytics.

2022 | Derived Fact Orchestration System (DFOS) - Amazon Worldwide Sustainability

In 2022, I led the design and development of the Derived Fact Orchestration System (DFOS), a core building block in Amazon's sustainability pipeline. DFOS automates the generation of derived emission factors from source models and datasets, replacing a manual and error-prone workflow with a production system that updates outputs within a day of upstream change. I provided end-to-end technical leadership across six major components, from onboarding and configuration to reactive computation and secure storage. The platform improved reliability, auditability, and data provenance, eliminated roughly two FTE days per month of manual work, and enabled more timely sustainability decision-making across finance, transportation, operations, and related domains.

2023 | AppSec Certification and Security Leadership - Amazon Worldwide Sustainability

In 2023, I led the first AppSec certification of the Sapphire environmental assessment stack within Amazon's Worldwide Sustainability organization. This year-long effort covered six core systems and required coordinated resolution of more than 50 baseline security controls and over 20 penetration-test findings. I defined workstreams, drove threat modeling across eight systems, authored security documentation, and institutionalized secure development practices later adopted by more than 150 engineers. This program unlocked secure onboarding of AWS and environmental-impact-factor data, elevated the security bar for sustainability engineering, and positioned the organization for broader internal adoption. I later became an AWS-certified Security Guardian and founded the AppSec Samurai group to scale review efficiency and security consistency across teams.

2024 | EIFMap - Amazon's First Generative AI System for Environmental Impact Factor Mapping

In 2024, I architected and led the end-to-end development of EIFMap, Amazon WWS's first generative AI system for environmental impact factor mapping. Built with AWS Bedrock, OpenSearch vector search, and a retrieval-augmented generation architecture, EIFMap automates a previously research-intensive process for mapping business activities to emission factors. I led the technical exploration required to make LLM-driven outputs trustworthy in production, including prompt chaining, validation mechanisms, grounded retrieval, and structured output handling. The system supported more than 100,000 mappings in its first six months, reduced mapping time and cost by over 90 percent, and helped avoid an estimated \$2 million in annual expense. The work also informed subsequent GenAI efforts across WWS and supported peer-reviewed dissemination through Environmental Science & Technology, NeurIPS 2024, and ISIE 2025.

Late 2024 - Present | Bomblehive - Modular Life Cycle Assessment Framework

Since late 2024, I have led the design and development of Bomblehive, a modular life cycle assessment framework that consolidates previously fragmented LCA implementations across Amazon and AWS. I re-architected a 17,000+ line legacy codebase into a testable, maintainable library with strong unit and integration coverage, centralized data dependency management, and reproducible enrichment pipelines spanning more than 30,000 emission factors and 19 background datasets. The design was reviewed and approved by Principal Engineers and required extensive alignment across science and engineering stakeholders. Bomblehive now underpins product-level environmental disclosures, reduces duplicated modeling effort, and creates a durable foundation for scalable, service-oriented environmental assessment.

2016-2019 | Competitive Price Recommendation - Amazon Seller Partners Pricing

Earlier at Amazon, I led one of the critical migrations in Seller Partners Pricing: the Competitive Price Recommendation system, a Tier-1 service that influenced seller pricing and Featured Offer outcomes. The legacy system lacked requirements documentation and depended on cross-time-zone coordination, so I reverse-engineered the behavior, designed a generic rule-engine approach to handle more than 10 rules and 30+ inputs, and implemented a shadow-testing framework to validate behavior before launch. The system operated in an environment driven by high-frequency price notifications and workloads around 40,000 transactions per second. The eventual launch was bug-free, generated no Sev1 or high-impact tickets, and unblocked multiple downstream roadmap features.

2010-2014 | Application Security Portfolio - Infosys Research Labs

At Infosys Research Labs, I helped develop novel application-security techniques that resulted in five granted U.S. patents covering tokenization, structure-preserving data masking, web content protection, offline DRM access, and electronic signature verification. Four of these patents list me as the primary or second inventor. The work began as research prototypes and was later productionized into enterprise platforms. These inventions addressed secure access to digital content and sensitive data across web, offline, and enterprise environments and helped establish formative approaches in application security. The portfolio was recognized internally with innovation awards and has since received citation attention from major industry players.

Granted U.S. Patents

Patent No.	Title	Granted	Rank	Cites
US9953171B2	System and method for tokenization of data for privacy	2018	2	18
US9805019B2	System and method for universal structure preserving data masking	2017	2	3
US10375210B2	Method and architecture for accessing digitally protected web content	2019	1	4
US9805350B2	System and method for providing access of digital contents to offline DRM users	2017	3	24
US9654295B2	System and method for verifying varied electronic signature	2017	2	2

Publications, Conference Papers, and Technical Guidance

Work	Venue / Status	Role
Parakeet: Emission factor recommendation for carbon footprinting with generative AI	NeurIPS 2024 Environmental Science & Technology ISIE 2025	Co-author
Algorithmic approaches to scale life cycle assessments	ISSST 2025 ISIE 2025	Contributor
Environmental Impact Factor Mapping on AWS	AWS technical guidance	Technical reviewer

Professional Memberships, Advisory, Judging, and Mentoring

- Fellow, The Institution of Electronics and Telecommunication Engineers (IETE)
- International Advisory Committee member, ICICC-2025
- Judge for Business Intelligence Group awards, including the Fortress Cybersecurity Awards and the Excellence in Customer Service Awards
- Judge for technical projects at a Generative AI-themed Major League Hacking (MLH) hackathon
- Mentor through MentorCruise, with a focus on application security, sustainability engineering, AWS architecture, and technical leadership

Technical Articles and Media Contributions

- Published technical articles in Dataconomy on the intersection of AI and application security
- Published technical articles in Cybersecurity News on digital rights management and emerging cybersecurity challenges
- Published technical articles in HackRead on next-generation AI-powered approaches to content protection and DRM

Awards and Distinctions

- Award for Innovation for research contributions at Infosys Ltd.
- Spot Award for valuable contributions at Infosys Ltd.
- Academic Excellence Award for the highest aggregate in the Computer Science undergraduate batch
- Graduated with a 4.0 / 4.0 GPA in the Master of Science program in Computer Science at the University of Southern California